

THARUN V PUTHANVEETIL

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- Senior Robotics Researcher with 5+ years of experience building perception, manipulation, and learning systems for real robots.
- Focused on human-in-the-loop vision, control, and data pipelines for closed-loop robotic autonomy.
- Published researcher with 100+ citations and 3 patents (filed/pending).

EDUCATION

University of Maryland - CP, USA | Master of Engineering in Robotics (GPA: 3.9/4.0)

Aug. 2022 – May 2024

Vellore Institute of Technology, India | B.Tech. in Electronics and Communication (GPA: 8.59/10.0)

Jul. 2014 – Apr. 2018

EXPERIENCE

Robotics Engineer | Deformable Object Manipulation, Robot Learning, Multimodal Perception

Jun. 2024 – Present

General Motors R&D

Michigan, USA

- Research on **Deformable Object Manipulation** with an emphasis on perception–control coupling and policy generalization.
- Designed closed-loop robotic pipelines integrating multimodal perception, feedback control, simulation, and learning.
- Built a modular ROS 2 perception system exposing detection, segmentation, and tracking as task-conditioned services.
- Integrated **Foundation Models** for perception and pose-driven servoring within ROS 2 framework.
- Reduced end-to-end perception latency by ~35% through Embedding reuse and temporal mask propagation.
- Sustained 30–40 FPS multi-object perception under concurrent tracking and segmentation workloads.
- Developed bimanual wire-harness manipulation pipelines using **Movelt 2** and **Impedance-based control**.
- Regulated elastic deformation by enforcing bounded tension using force–torque feedback at the end-effectors.
- Observed a 2× increase in manipulation failure rate when tension feedback was disabled under dynamic routing.
- Constructed **high-fidelity Digital Twins** modeling dynamics, sensing noise, and actuation delays.
- Employed Digital Twins to study perception and control behavior under varied simulated task conditions.
- Designed **Sim2Real** transfer pipelines to study performance degradation under domain shift and unmodeled dynamics.
- Automated large-scale synthetic data generation in Isaac Sim using **Behavior Trees** for task diversity.
- Built **Human-in-the-Loop** teleoperation systems for collecting learning-grade demonstrations in Joint & Cartesian spaces.
- Recorded synchronized multimodal datasets using **Movelt Servo** with raw MCAP/ROS bag logging.
- Improved object-detection robustness by **20–30%** using **hard-negative sampling** and **structured distractor generation**.

Researcher (AI & Robotics) | Multi-Modal Networks, Data Security, Action Recognition

Feb 2023 – Mar 2024

CATT Labs

Maryland, USA

- Developed **Multi-Modal Anomaly Detectors** to report behaviors that deviate from the robot's normal baseline nature.
- Created a one-stop log retrieval service using **Large Language Models** to improve accessibility for non-technical users.
- Formulated **metrics** to measure, compare, and parameterize the safety of robots and cyber-physical systems.

Track Anything Rapter (TAR) | Python, PyTorch, OpenCV, ROS2 [[github](#)] [[preprint](#)]

- Realized a UAV system to achieve **one-shot object tracking** using **Foundation Models** from multi-modal queries.
- Implemented a custom **ROS2 Gazebo** simulation pipeline for validation of tracking algorithms.
- Developed a **proportional-based visual servoing controller** leveraging **PX4 Autopilot** for closed-loop object tracking.
- Transitioned to a **ROS2-based** control system utilizing **MAVROS**, bypassing a dependency on **MAVSDK**.
- Achieved a mean **Dynamic Time Warping (DTW)** score of 1.68 for UAV trajectory against **Vicon** ground truth.

Fall Detection using Human Pose Estimation | Python, PyTorch, OpenCV [[github1](#)] [[github2](#)] [[preprint](#)]

- Crafted a **Human Pose Estimation**-keypoint-based fall detection model; identified optimal framerate for fall events.
- Quantified the potency of each **pose keypoint** and the relevance of **camera angle** in improving accuracy.
- Custom **GCN + Transformer** model achieved 95% (avg.) accuracy on both UR & NTU datasets.
- Devised a pipeline for multi-view-based pose aggregation for fall recognition in **partial/no occlusion** scenarios.
- **GCN+Transformer** multi-view aggregation model achieved 96% accuracy for multi-class action recognition.

Researcher (AI & Robotics) | Human-Robot Interaction, Computer Vision, Precision Manipulation

Jun. 2021 – Jan. 2023

Indian Institute of Science & Perception and Robotics Group, UMD

Bangalore, India & Maryland, USA

- Implemented an **Imitation Learning-based policy** to instruct robot actions from human demonstration, speech & gestures.
- Constructed a **U-Net-based Monocular Depth Estimation** model on the NYU dataset with 94% accuracy (Top 4 @ IROS21).

Autonomous Weeding Robot | Python, Matlab, OpenCV, TensorFlow, PyTorch, ROS [[video](#)] [[github](#)]

- Engineered a **robotic manipulator**-based plucking robot to reduce pesticide usage in indoor farms by up to 80%. [[Report - Link](#)]
- Created a **YoloR** object detector trained on augmented synthetic data for weed localization with 98% accuracy.
- Implemented an IK solver based on supervised **Behavioral Cloning** for precise manipulation.
- Developed a precision weeding gripper with adaptive vision-based control for minimal non-weed contact. [[Patent - Link](#)]
- Designed an **adaptive PID controller** to enable weeding while minimizing disturbance to the plant bed.

Project Engineer | HCI, Photogrammetry, Recommendation Engine, Pattern Recognition

Jul. 2018 – May 2021

CTO Office, Wipro Digital

Bangalore, India

- Led as the **AI Project Lead** at the Innovation team, orchestrating the quarterly project lineup.
- Developed a **2D CNN**-based tool tracking system using Leap Motion controller, enhancing hardware assembly pipelines.
- Engineered a mobile-centric insurance solution with a **Transfer Learning**-based car model detector (96% acc.) in **TFLite**.
- Implemented **Generative AI** and learning-based algorithms for **my Style**, one of Wipro's top retail POCs (2020).

my Style | Python, OpenCV, TensorFlow, PyTorch, MySQL, Flask [[link1](#)] [[link2](#)] [[video](#)]

- End-to-end AI-powered shopping app to reduce trial room drops and improve scope for apparel customization.
- Implemented body measurement extraction via **2D photogrammetry & Human Pose Estimation**.
- Developed a **RASA-based chatbot** for personalized dress recommendations.
- Built a **GAN**-based product customization model for dynamic apparel styling (82% higher customer traction).
- Engineered a supervised **Fit Analyzer** model to evaluate apparel fit as a 'Fit%' metric.

PUBLICATIONS/PATENTS/POSTERS

[1] ASTROD: Multimodal Anomaly Detection for Autonomous CPS Empowering Real-World Evaluation. GameSec 2024. [\[Link\]](#)

[2] A Data-Driven Approach for Securing Autonomous Cyber-Physical Systems. CDC 2025. [\[Manuscript\]](#)

[3] Application of Mobile Collaborative Robot using Deep Learning in Precision Weed Control. Elsevier 2021. [\[Manuscript\]](#)

[4] Event-based Dynamic Obstacle Avoidance in Outdoor Environments. IROS 2021. [\[Poster\]](#)

[5] A univariate data analysis approach for rainfall forecasting. ICCIS 2020. [\[Link\]](#)

[6] Prediction of rainfall using data mining techniques. ICICCT 2018. [\[Link\]](#)

[7] A comparative study of various classification techniques to determine water quality. ICICCT 2018. [\[Link\]](#)

[8] Wall climbing robot using soft robotics. ICPCSI 2017. [\[Link\]](#)

TECHNICAL STACK

Programming: Python, C/C++, SQL (MySQL), JavaScript, HTML/CSS, Matlab, LabVIEW, Docker

AI/ML/CV/Data: TensorFlow, PyTorch, OpenCV, Keras, scikit-learn, Tableau, LangChain, Pydantic, OpenAI, NLTK

Embedded/IoT/Robotics/Sim: ROS, ROS2, MQTT, HTTP, REST, Sockets, Soft Robotics, Unity, Flask, Gazebo, RViz

Robotic Platforms: Turtlebot 2/3, Delta, OpenManipulator-X, Husky, M500 Drone, UR5/UR5e, UR10

S/W Development: Agile, Jira, Git/GitLab/Bitbucket, Pair Programming, TDD, UML, SCM

ADDITIONAL PROJECTS

AI | COMPUTER VISION | MACHINE LEARNING

Image Segmentation using SLIC <i>Python, OpenCV, TensorFlow, HTML, CSS, Angular.js, Unity</i> [link]	Nov. 2023 – Dec. 2023
- Utilized Simple Linear Iterative Clustering (SLIC) to generate superpixels for a custom segmentation dataset. - Implemented a CNN-based deep feature extractor for superpixel-based image segmentation. - Achieved 20% improved segmentation performance over k-NN-based segmentation.	
Image Outpainting <i>Python, OpenCV, TensorFlow, HTML, CSS, Angular.js, Unity</i> [link]	Nov. 2023 – Dec. 2023
- Implemented Image Outpainting using Implicit Neural Representation. - Observed ~3-point lower PSNR using Positional Encoding add-on vs. FFN-only baseline.	
AI Cricket Coach <i>Python, OpenCV, TensorFlow, HTML, CSS, Angular.js, Unity</i> [link]	Jul. 2020 – Present
- Classifies bowl types from video feeds; provides coaching assistance in a WebVR environment. - Created a Body Pose Estimation plugin for batting pose feedback. - Custom 3D CNN bowl classifier (92% accuracy) trained on synthetic VR data.	
Virtual Try-On <i>Python, OpenCV, TensorFlow, PyTorch, Flask</i> [link]	Jan. 2021 – Mar. 2021
- AI solution to reduce trial room drops and provide a style quotient evaluator. - Built a GAN-based texture generator rendered onto 3D AR apparel models. - Developed an Aesthetic Quality Assessment Model (AQAM) and an Explainable AI plugin.	

ROBOTICS

Maze Runner <i>C++, ROS2, Gazebo, RViz</i> [link1] [link2]	Nov. 2023 – Dec. 2023
- Autonomous maze navigation using dynamic cues from ArUco markers in Gazebo. - Implemented a ROS2 Action-Client that uses Nav2 for visual feed-based waypoint navigation.	
Leonardo - Autonomous Retrieval UGV <i>Python, Arduino, OpenCV</i> [video]	Feb. 2023 – May 2023
- Constructed a Barron robot with IMU, encoder, RPi camera, and a servo-based arm. Visual Servoing-based low-level controller for object pick-and-place with onboard arm. - Rapid localization via sensor fusion of IMU, encoder, & range sensor in confined zones.	
Auto Platoon <i>Python, OpenCV, PyTorch</i> [video] [link]	Apr. 2023 – May 2023
- Bio-inspired multi-agent leader-follower system for modular & energy-efficient transport. YOLOv7+Kalman-based agent tracking with dynamic obstacle avoidance. - Socket-based connected-vehicle communication and Visual Servoing-based motion planning.	
ACO-RRT* – Bio-Inspired Path Planning <i>Python, NumPy</i> [link]	May 2023 – Jun. 2023
- Implemented a bio-inspired planner that explores random trees via ant foraging behavior. - Improved traditional RRT* sampling with Ant Colony Optimization. - Achieved 1.4× and 3.54× faster convergence (vs. RRT*) for first and ideal paths, respectively.	

VOLUNTEERING EXPERIENCE

Electrical Programming Head Coach Robotics, Software Development, Computer Vision	Jun. 2024 – Present
<i>The Irrational Engineers</i>	<i>Maryland, USA</i>
Mentoring diverse high school students in building competitive robots for FIRST Robotics, focusing on software for Controls and Computer Vision.	

GCN - Graph Convolution Network; GAN - Generative Adversarial Network; TSA - Time Series Analysis; SCM - Software Configuration Management; UGV - Unmanned Ground Vehicle.